PRELIMINARY WORK ITEM International Standard

ISO/PWI 25534-1

ISO/JTC **5**

Secretariat: **DIN**

Digital product passport — Vocabulary

Passeport numérique de produit — Vocabulaire

ICS: 35.240.63

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Foreword

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This document was prepared by Joint Technical Committee ISO/IEC JTC 5, *Digital product passport* (secretariat DIN).

This preliminary work item draft was prepared by CalConnect (The Calendaring and Scheduling Consortium), Category A liaison applicant to ISO/IEC JTC 5, as an input document for the proposed terminology work item arising from the proposed division of ISO 25534-1 into a Part 1, Vocabulary, and a Part 2, Overview and fundamental principles. It is derived from a machine-readable concept dataset of 145 verified concepts published at https://glossarist.org/dpp/ (mirror: https://unidpp.org/terminology/). It does not represent a consensus of ISO/IEC JTC 5.

Introduction

Two incompatible systems of definitions for the digital product passport (DPP) are being developed in parallel, and they do not cite each other normatively.

The eight European Standards of the EN 182xx series (prepared by CEN-CLC/JTC 24 under the European Commission's standardisation request M/604 in support of the Ecodesign for Sustainable Products Regulation) define the digital product passport as a “digital record of product characteristics throughout its life cycle”. The working draft ISO/AWI 25534-1, under development in ISO/IEC JTC 5, defines it as “a concept or technology which enables sharing of product related information”. The core noun — a record, on the one hand; a concept or technology, on the other — is normatively incompatible between the two systems that are required to interoperate. The divergence is not confined to the head term. Full-text comparison of the corpus (the eight EN 182xx standards, CWA 18291:2025, the Asset Administration Shell specification IDTA 01001-3.2:2026, and ISO/AWI 25534-1) shows casing drift between implementations of the same field (`lastUpdated` in EN 18223:2026 against `lastUpdate` in IDTA 02099-1), three unreconciled lifecycle taxonomies inside the EN series itself (the identifier-granularity model of EN 18219:2026, the nine-phase model of EN 18239:2026 Annex A, and the ISO 14040:2006 lineage used by other parts of the series), three lookup systems (the resolver of EN 18219:2026, the registry API of EN 18222:2026 and the central EU registry) whose relationships are never specified, and role models that are never mapped to the API-level authorisation that would enforce them. None of these divergences is a mere editorial blemish: each is a point at which two implementers, each conforming to the letter of a different document, produce systems that do not interoperate. A common vocabulary is the prerequisite for repairing them.

This document addresses that prerequisite. It presents a vocabulary for the fundamentals of the digital product passport, compiled by the following method:

— the terms and definitions clauses of the corpus were harvested in full — 141 term entries (82 from the EN 182xx series, 30 from CWA 18291:2025 and 29 from IDTA 01001-3.2:2026), together with the terms clause of ISO/AWI 25534-1 (revision of 2026-09-05);

— the harvest was verified, merged and annotated concept by concept into a machine-readable concept dataset of 145 concepts, each carrying its definition, its designations and its source citations, with equivalences and conflicts recorded per concept;

— this working draft converts that dataset into terminological entries: the core vocabulary clauses (3.1 to 3.7) contain 124 entries aligned, where an international source exists, to that international source and citing the EN/CWA formulations as regional variants; the framework clauses (3.8 to 3.16) define 99 entries for the projection, provenance, tier and trust model that the corpus does not name.

The framework terms merit explanation, because they are not drawn from the corpus — the corpus cannot supply them. The analysed European Standards model the passport as a static serialized document: their class taxonomy classifies JSON shapes rather than meanings, their semantic repository is a placeholder (EN 18223:2026 requires a definition repository but does not specify one), and their own conflict-resolution rule — that prose is authoritative where diagrams, text and JSON disagree — records the absence of a formal model. The proposed framework architecture supplies the missing concepts:

— **The projection doctrine.** The passport is the legal projection of the digital twin: an event-sourced, append-only record of testimonies about the subject is the source of truth, and the passport is a deterministic, registered, legally cited view over it. A notarized “as-of” reconstruction answers what was legally true at a specified past time. The corpus has no twin concept, no provenance classes, no precedence rules and no time model; the framework terms of 3.9 to 3.11 name these.

— **The lens model.** A profile is a calibrated lens placed on the twin: a versioned, registered projection function that selects and transforms — granularity as focal length, traversal depth as depth of field, classification as colour filter, unit conversion as filter factor, published uncertainty as a mandatory property. Profiles constrain and transform; they never redefine (invariant of the framework). Inter-profile disagreement between jurisdictions is expected, published as decision rules, and auditable — never hidden and never silently merged. The standardized interface by which any registered profile attaches to the core — the lens mount — is the interoperation guarantee.

— **The degradation ladder.** Service is organized in tiers: a carrier-embedded minimum (Tier A) verifiable offline with pre-cached trust anchors, a served full passport (Tier B), and a notarized archive (Tier C). Degradation is explicit: freshness verdicts mark staleness visibly; the system degrades to the document-lookup reality of today's deployments without losing verifiability for those who need more.

— **The trust model.** Trust is graded, not binary: every element carries a trust marker; verdicts state which of three legally distinct readings they answer (evidentiary, current-state, cryptographic) and carry a coverage report; jurisdictional trust lists cross-recognize through a multi-witnessed master list; and the reason for a revocation determines its retroactivity. Enumeration resistance — the property that the system is verifiable without being browsable — protects the installed base of every participant.

These four commitments — projection, lenses, tiers, graded trust — generate the framework vocabulary. The terms are defined here so that the committee can evaluate the architecture through its vocabulary, term by term, rather than as an indivisible proposal. Each framework entry is marked by its placement in 3.8 to 3.16 and carries no external source citation; the sources-and-harmonization matrix in Annex A records, for every other entry, which source standard defined it and where definitions diverged.

This document is intended to support the division of ISO 25534-1 into a Part 1, Vocabulary, and a Part 2, Overview and fundamental principles, and to provide seed content for the semantic definition repository that EN 18223:2026 requires but does not specify.

Digital product passport — Vocabulary

1 Scope

This document establishes a vocabulary for the digital product passport (DPP).

It defines terms for the fundamental concepts of the digital product passport as used in the documents that currently govern or inform its implementation, and it harmonizes those concepts where their definitions conflict. The source corpus comprises the EN 182xx series (EN 18216:2026 to EN 18223:2026, EN 18239:2026 and EN 18246:2026), CWA 18291:2025, the Asset Administration Shell metamodel specification IDTA 01001-3.2:2026, and ISO/AWI 25534-1, together with the further standards cited as sources of individual definitions.

In addition to the harmonized terms (3.1 to 3.7), this document defines terms for a framework architecture of projections, profiles, service tiers and graded trust (3.8 to 3.16), developed to name concepts that the source corpus requires but does not define, including the digital twin as the subject of the passport, the provenance and precedence of facts, the composition and transformation of passports, the capability classes of subjects, and the verification semantics of trust.

This document supports the proposed division of ISO 25534-1 into a Part 1, Vocabulary, and a Part 2, Overview and fundamental principles. The terms defined here are intended to serve all subsequent parts of ISO 25534 and documents referring to them.

2 Normative references

There are no normative references in this document.

3 Terms related to the digital product passport

ISO and IEC maintain terminology databases for use in standardization at the following addresses:

— ISO Online browsing platform: available at https://www.iso.org/obp

— IEC Electropedia: available at https://www.electropedia.org

3.1
digital product passport
concept or technology which enables sharing of product related information that is essential for product sustainability and circularity, including those specified in regulations, across all relevant economic actors

Note 1 to entry: Example characteristics include environmental sustainability, environmental impact, and recyclability.

Note 2 to entry: Definitional conflict: the EN 182xx series (18216, 18221, 18222, 18223, 18239, 18246) define DPP as a “digital record of product characteristics throughout its life cycle” — differing in noun (concept/technology vs record) and in stated scope (across relevant economic actors vs throughout the life cycle).

Note 3 to entry: Definitional variant: EN 18219:2026 (§3.1.3) and EN 18220:2026 (§3.6) extend the EN-family definition with “accessible via electronic means through a data carrier” — a functional statement of the carrier subset of the ISO concept.

[SOURCE: CEN EN 18220:2026, 3.6[56] , modified — modified, CEN EN 18223:2026, 3.4, modified, CEN EN 18216:2026, 3.7, modified, CEN EN 18221:2026, 3.2, modified, CEN EN 18222:2026, 3.2, modified, CEN EN 18239:2026, 3.9, modified, CEN EN 18246:2026, 3.2, modified, CEN EN 18219:2026, 3.1.3, modified, ISO/AWI 25534-1:2026, 3.1]

3.2
digital product passport id
unique identifier of an instance of the digital product passport

[SOURCE: CEN EN 18223:2026, 3.5[59]]

4 Terms related to identity and carriers

4.1
unique product identifier
unique string of characters for the identification of a product that also enables a web link to the digital product passport

[SOURCE: CEN EN 18246:2026, 3.3[61] , modified]

4.2
unique identifier
identifier which is guaranteed to be unique among all identifiers used for those objects and for a specific purpose

Note 1 to entry: A unique identifier refers to unique product identifier (*unique product identifier* (4.1)), unique economic operator identifier (*unique economic operator identifier* (4.12)), and unique facility identifier (*unique facility identifier* (4.13)).

Note 2 to entry: For more information, see ISO/IEC 15459-1.

Note 3 to entry: Regional variant: the definition recorded from the merged concept pair was: string of characters that is required to be unique among all identifiers used for all objects for a specific purpose

Note 4 to entry: Recorded variance: EN 18219:2026 modification: Note 1 to entry removed and a new Note 1 to entry added. EN 18220:2026 modification: “identifier” replaced by “string of characters”, Note 1 to entry removed and a new Note 1 to entry added.

[SOURCE: ISO 29404:2015, 3.26, modified — CEN EN 18219:2026, 3.1.24, similar, CEN EN 18220:2026, 3.18, similar, CEN CWA 18291:2025, 3.3, ISO/IEC 15459-1]

4.3
digital identifier
sequence of characters associated with digital, non-digital, or abstract entities, such as books, images, reports, metadata records or events

[SOURCE: ISO 24619:2011, 3.2.1, modified — CEN EN 18216:2026, 3.1]

4.4
identification
process of recognizing an object in a particular domain as distinct from other objects

[SOURCE: ISO/IEC 24760-1, 3.2.1, modified — CEN EN 18216:2026, 3.3]

4.5
identification scheme
system for allocating identifiers to objects

[SOURCE: ISO/IEC 6523-1, 3.6, modified — modified, CEN EN 18219:2026, 3.1.6]

4.6
domain
specified area of knowledge, application, or activity within which a unique identifier is used

[SOURCE: CEN EN 18219:2026, 3.1.4[55]]

4.7
self-issuing system
decentralized identification scheme
system or mechanism that an organization uses to generate and assign unique identifiers to its objects without the intervention or oversight of an external authority

[SOURCE: CEN EN 18219:2026, 3.1.20[55]]

4.8
persistence
process to keep a unique identifier associated with an identified object throughout its life cycle

[SOURCE: CEN EN 18219:2026, 3.1.13[55] , modified — CEN EN 18220:2026, 3.12]

4.9
preservation
activity of ensuring that a unique identifier and its denotation remain stable, valid, and retrievable for as long as required

[SOURCE: CEN EN 18219:2026, 3.1.15[55]]

4.10
resolution
process in which an identifier is translated or mapped to another identifier, associated information or metadata about the object it represents

[SOURCE: CEN EN 18219:2026, 3.1.18[55] , modified]

4.11
resolver
system or service that performs resolution (*resolution* (4.10))

[SOURCE: CEN EN 18219:2026, 3.1.19[55] , modified]

4.12
unique economic operator identifier
unique string of characters for the identification of an actors involved in a product’s value chain

Note 1 to entry: The source prints “an actors”; the definition is reproduced verbatim.

[SOURCE: CEN EN 18219:2026, 3.1.22[55]]

4.13
unique facility identifier
unique string of characters for the identification of location or building involved in a product’s value chain or used by actors involved in a product’s value chain

[SOURCE: CEN EN 18219:2026, 3.1.23[55]]

4.14
unique registration identifier
identifier returned after uploading the unique identifiers and any other data required to the DPP registry

EXAMPLE Unique identifiers can be unique product identifier, unique facility identifier, unique operator identifier of the economic operator etc.

[SOURCE: CEN EN 18222:2026, 3.6[58]]

4.15
data carrier
device or medium used to store data as a relay mechanism in an automatic identification and data capture system

[SOURCE: CEN EN 18220:2026, 3.5[56]]

4.16
additional software
additional application, program, or tool that a user will install or access separately from the default setup of most smartphones or similar devices to interact with the digital product passport

EXAMPLE Proprietary applications, plugins or extensions, dedicated apps, specialized middleware, or customized software readers.

Note 1 to entry: Standard functionalities, such as web browsers, camera-based QR readers, and universal communication protocols (e.g. DNS, HTTP, HTTPS), are not considered as additional software.

[SOURCE: CEN EN 18220:2026, 3.1[56]]

4.17
barcode
printed data carrier indicating either a reference to a linear or a 2D matrix

Note 1 to entry: It excludes RFID data carriers.

[SOURCE: CEN EN 18220:2026, 3.2[56]]

4.18
near field communication
NFC
subset of HF passive radio-frequency identification with low range

[SOURCE: CEN EN 18220:2026, 3.11[56]]

4.19
radio-frequency identification
identification technology that uses electromagnetic fields to automatically identify and track tags attached to objects

[SOURCE: CEN EN 18220:2026, 3.15[56]]

4.20
RAIN RFID
UHF RFID
passive UHF radio-frequency identification (*radio-frequency identification* (4.19))

Note 1 to entry: Source entry heading: “RAIN RFID (UHF RFID) per ISO/IEC 18000-63”.

[SOURCE: CEN EN 18220:2026, 3.16[56]]

4.21
two-dimensional symbol
code representing data in machine-readable form by a collection of polygonal or circular cells in a regular pattern which are read optically by scanning

[SOURCE: CEN EN 18220:2026, 3.17[56]]

4.22
X dimension
specified width of the narrow in a bar code symbol or the specified width of a single element in a two-dimensional symbol

[SOURCE: ISO/IEC 19762, 3.2.1.10, modified — CEN EN 18220:2026, 3.19]

4.23
batch
subset of a model (*model* (4.25)) that is grouped by an economic operator based on common characteristics

[SOURCE: CEN EN 18220:2026, 3.3[56] , modified — similar, CEN EN 18219:2026, 3.1.1]

4.24
item
single unit of a model (*model* (4.25))

[SOURCE: CEN EN 18220:2026, 3.8[56] , modified — similar, CEN EN 18219:2026, 3.1.9]

4.25
model
version of a product of which all units share the same technical characteristics and the same model identifier

[SOURCE: CEN EN 18219:2026, 3.1.11[55] , modified — CEN EN 18220:2026, 3.10]

4.26
identifier
identity information that unambiguously distinguishes one entity from another one in a given domain

Note 1 to entry: There are specific identifiers, e.g. UUID Universal unique identifier, IEC 15418 (GS1).

[SOURCE: Glossar Industrie 4.0[47]]

5 Terms related to actors and issuance

5.1
economic operator
manufacturer, authorized representative, importer, distributor, dealer or fulfilment service provider

Note 1 to entry: Regional variant: the definition recorded from the merged concept pair was: manufacturer, authorized representative, importer, distributor, dealer or fulfilment service provider that places the product on the market

[SOURCE: CEN EN 18219:2026, 3.1.5[55] , modified — similar, CEN EN 18220:2026, 3.7, similar, CEN EN 18221:2026, 3.1]

5.2
digital product passport service provider
natural or legal person that is an independent third party authorized by the economic operator (*economic operator* (5.1)) that places the product on the market or puts it into service and that processes the digital product passport data for that product for the purpose of making such data available to economic operators and other relevant actors with a right to access those data

[SOURCE: CEN EN 18221:2026, 3.3[57]]

5.3
main digital product passport service provider
digital product passport service provider (*digital product passport service provider* (5.2)) that is hosting the digital product passport (*digital product passport* (3.1)) on behalf of the economic operator (*economic operator* (5.1)) if the economic operator chooses not to host the digital product passport themselves

[SOURCE: CEN EN 18221:2026, 3.3.1[57]]

5.4
back-up digital product passport service provider
digital product passport service provider (*digital product passport service provider* (5.2)) that is hosting the back-up copy of the digital product passport (*digital product passport* (3.1))

[SOURCE: CEN EN 18221:2026, 3.3.2[57]]

5.5
issuing agency
organisation entrusted by a Registration Authority (RA) to manage unique identification schemes

[SOURCE: CEN EN 18219:2026, 3.1.8[55]]

5.6
actor
organization or individual that fulfils a role

[SOURCE: ISO 23234:2021, 3.4, modified — CEN EN 18239:2026, 3.7]

5.7
notified actor
organization or individual entitled by an authorized accrediting body or authority, that fulfils a role in the DPP lifecycle

[SOURCE: CEN EN 18239:2026, 3.8[60]]

5.8
consumer
individual member of the general public purchasing or using goods, property or services for private purposes

[SOURCE: CEN EN ISO 14025:2010, 3.16[62] , modified — CEN EN 18219:2026, 3.1.2, CEN EN 18220:2026, 3.4, CEN CWA 18291:2025, 3.24]

5.9
customer
organization or person that receives a product

Note 1 to entry: The customer may be the user or a distributor [SOURCE: ISO 9000].

[SOURCE: ISO 9000:2015]

5.10
expert
expert in charge of performing a study

[SOURCE: CEN CWA 18291:2025, 3.8[53]]

5.11
reviewer
expert in charge of reviewing a study

[SOURCE: CEN CWA 18291:2025, 3.9[53]]

5.12
exporter
name and address of the person who makes, or on whose behalf the “export declaration” is made, and who is the owner of the goods or has similar right of disposal over them at the time when the declaration is accepted

[SOURCE: ISO/TS 24533:2012]

5.13
seller
company or company function that sells a product

[SOURCE: CEN CWA 18291:2025, 3.11[53]]

5.14
subcontractor
company or company function that performs activities as a sub-contractor (for example cutting/sewing operations but also dyeing)

[SOURCE: CEN CWA 18291:2025, 3.12[53]]

5.15
buyer
individual or entity purchasing goods or services

[SOURCE: ISO 24533-2:2022]

5.16
producer
company or company function that produces final or non-final fashion products (yarns, fabrics, garments, accessories, footwear, etc.); its industrial process has in input materials and in output components or final fashion goods for the business customers or final customers

[SOURCE: CEN CWA 18291:2025, 3.14[53]]

5.17
supplier
company or company function that produces non final products (accessories, fabrics, yarns…); its industrial process has in input materials and in output components for the business customers

[SOURCE: CEN CWA 18291:2025, 3.15[53]]

5.18
yarn producer
company or company function that produces yarns from raw material; it has mainly Fabric and Knitwear producers as customers (Apparel Producers sometimes)

[SOURCE: CEN CWA 18291:2025, 3.16[53]]

5.19
retailer
company or company function that sells garments to the Consumer. It is a client of Apparel Producer, Knitwear Producer, and Yarn Producer

[SOURCE: CEN CWA 18291:2025, 3.17[53]]

5.20
apparel producer
company or company function that organizes manufacturing and material supplying to obtain finished garments from fabrics. It is a client of Fabric Producer and partially of Yarn Producer

[SOURCE: CEN CWA 18291:2025, 3.18[53]]

5.21
sorter
person or a system that categorizes discarded textile materials by their fibre composition, color, and condition to determine their potential for reuse, repair, or recycling

[SOURCE: CEN CWA 18291:2025, 3.19[53]]

5.22
textile collector
actor involved in separate collection of textiles

[SOURCE: ISO 5157:2023]

5.23
recycler
company or company function that performs activities to obtain recovered resources for use in a process or a product, excluding energy recovery

Note 1 to entry: Activities to obtain resources include activities such as recovery, collection, transport, sorting, cleaning, and re-processing.

Note 2 to entry: Recycling does not include reuse.

[SOURCE: ISO 59004:2024]

5.24
eService platform
platform for eBusiness centralized services

[SOURCE: CEN CWA 18291:2025, 3.5[53]]

5.25
traceability platform
platform dedicated to traceability services

[SOURCE: CEN CWA 18291:2025, 3.6[53]]

5.26
sustainability management platform
platform dedicated to sustainability data collection, management and reporting

[SOURCE: CEN CWA 18291:2025, 3.7[53]]

6 Terms related to lifecycle and events

6.1
life cycle
consecutive and interlinked stages of a product system, from raw material acquisition or generation from natural resources to final disposal

[SOURCE: ISO 14040:2006, 3.1, modified — ISO/AWI 25534-1:2026, 3.2, CEN EN 18219:2026, 3.1.10, CEN EN 18220:2026, 3.9]

6.2
product
any physical good that is placed on the market (*placed on the market* (6.3)) or put into service (*put into service* (6.4))

[SOURCE: CEN EN 18220:2026, 3.14[56] , modified — similar, CEN EN 18219:2026, 3.1.16]

6.3
placed on the market
first making available of a product

[SOURCE: CEN EN 18219:2026, 3.1.14[55]]

6.4
put into service
first use, for its intended purpose, of a product

[SOURCE: CEN EN 18219:2026, 3.1.17[55]]

6.5
digital product passport lifetime
period during which the digital product passport (*digital product passport* (3.1)) is required by regulation to be available to any operator or natural or legal person with a right to access its data

[SOURCE: CEN EN 18221:2026, 3.4[57]]

6.6
end-of-use
point in time during the life cycle at which a product or resource is transferred by the holder to some other holder

EXAMPLE When an organization (e.g., shop or retailer) in possession of the product considers it to be a waste since it hasn't or can't be sold.

[SOURCE: ISO 59004:2024]

6.7
end-of-life
point in time during the life cycle at which a product or resource is taken out of use and is disposed

Note 1 to entry: Disposal could be in the form of incineration, deposit to landfill, or composting.

[SOURCE: ISO 59004:2024]

6.8
collection
process of gathering and transporting used textile products or waste materials to a sorting facility for further processing such as reuse, repair, remanufacturing, or recycling

[SOURCE: ISO 5157:2023]

6.9
closed loop system
system by which products or resources are used and then recovered and turned into new products or recovered resources, without losing their inherent properties

[SOURCE: ISO 59004:2024]

6.10
traceability
ability to trace the history, application or location of an object in a value chain

Note 1 to entry: Lineage: ISO/AWI 25534-1, 3.5 adopts this definition from ISO 9000:2015, modified by the addition of "in a value chain".

[SOURCE: UNECE Recommendation No. 46[49] , modified — CEN CWA 18291:2025, 3.29]

6.11
traceability system
practical system of processes, procedures and information exchanges that implements traceability

[SOURCE: UNECE Recommendation No. 46[49]]

6.12
traceable asset
traceable asset is any product or material (individually, in batches, or in trade units) that needs to be tracked along a value chain

[SOURCE: UNECE Recommendation No. 46[49]]

6.13
textile product
product made mainly of textile fibres, yarns and/or fabrics and intended to be used, as such or in conjunction with other textile or non-textile elements

Note 1 to entry: These articles can contain non-textile parts, such as plastics (e.g. buttons and membrane or coatings) or metals.

Note 2 to entry: Source entry heading: "textile product – product".

[SOURCE: ISO 5157:2023]

6.14
circular economy
products and materials are designed in such a way that they can be reused, remanufactured, recycled or recovered and thus maintained in the economy for as long as possible, along with the resources of which they are made, and the generation of waste, especially hazardous waste, is avoided or minimized and greenhouse gas emissions are prevented or reduced, can contribute significantly to sustainable consumption and production

[SOURCE: ISO/AWI 25534-1:2026, 3.3, modified]

6.15
product circularity
refers to repairing, reusing and remanufacturing equipment, and harvesting and recycling metals indefinitely through product design and collection processes

[SOURCE: ISO/AWI 25534-1:2026, 3.4, modified]

6.16
value chain
all activities and stakeholders that provide or receive value from designing, developing, making, distributing, retailing, and consuming a product (or providing the service that a product renders), including the extraction and supply of raw materials, as well as activities involving the product after its useful service life has ended

Note 1 to entry: Compared with the source: the source is framed for the textile sector; the AWI draft reformulates the term in a sector-neutral form.

[SOURCE: ISO/AWI 25534-1:2026, 3.8, modified — UNEP Job no. DTI/2532/PA, modified]

7 Terms related to trust and verification

7.1
controlled DPP data
information on digital product passport whose access is controlled based on the user's access rights

Note 1 to entry: User in the sense of a person who interacts with a system, product or service.

[SOURCE: CEN EN 18239:2026, 3.1[60] , modified — CEN EN 18216:2026, 3.8]

7.2
authentication
verification that a claimed identity is correct

[SOURCE: ISO/IEC/IEEE 8802-1AR, 3.2, modified — CEN EN 18216:2026, 3.4]

7.3
data integrity
property that data has not been altered or destroyed in an unauthorised manner

Note 1 to entry: In the context of secure communication, data integrity ensures that data transmitted between parties remains unaltered and intact from the moment it leaves the sender to the moment it reaches the receiver. This means that the data has not been tampered with, modified, or corrupted during transmission – whether accidentally or through malicious actions.

[SOURCE: ISO 7498-2:1989, 3.3.21, modified — modified, CEN EN 18216:2026, 3.5]

7.4
secure communication
mechanism of transmitting data between systems in a way that ensures its confidentiality, integrity and authenticity

[SOURCE: CEN EN 18216:2026, 3.6[54]]

7.5
system resilience
ability to recover from security compromises or attacks

[SOURCE: ISO/IEC 29180:2012]

7.6
system availability
property of being accessible and usable on demand by an authorized entity

[SOURCE: ISO/IEC 27000]

7.7
cyber resilience
ability to maintain business continuity despite adverse conditions, attacks, or compromises on critical data flow and related information systems

[SOURCE: CEN EN 18239:2026, 3.4[60]]

7.8
recovery point objective
RPO
point in time to which data must be recovered after a disruption has occurred

[SOURCE: ISO/IEC 27031, 3.12, modified — CEN EN 18239:2026, 3.5]

7.9
recovery time objective
RTO
period of time within which minimum levels of services and/or products and the supporting systems, applications, or functions must be recovered after a disruption has occurred

[SOURCE: ISO/IEC 27031]

7.10
object
entity to which specified requirements apply

EXAMPLE Product, process, service, system, installation, project, data, design, material, claim, person, body or organization, or any combination thereof.

[SOURCE: ISO/IEC 17000:2020, 4.2, modified — modified, CEN EN 18219:2026, 3.1.12]

7.11
PIA
RFID privacy impact assessment process
RFID Privacy Impact Assessment Process

[SOURCE: CEN EN 18220:2026, 3.13[56]]

7.12
claim
high-level statement about a characteristic of a product, or about a process or an organization associated with that product (traceable asset). A sustainability claim is a claim that covers one or multiple sustainability dimensions (economic, environmental, social)

Note 1 to entry: In this document “sustainability claims” are referred to as just “claims”.

[SOURCE: UNECE Recommendation No. 46[49]]

7.13
state-of-the-art
developed stage of technical capability at a given time as regards products, processes and services, based on the relevant consolidated findings of science, technology and experience

[SOURCE: ISO/IEC Guide 2:2004, 1.4, modified — CEN EN 18219:2026, 3.1.21]

7.14
access control
protection of system resources against unauthorized access; a process by which use of system resources is regulated according to a security policy and is permitted by only authorized entities (users, programs, processes, or other systems) according to that policy

[SOURCE: IEC TS 62443-1-1:2009]

7.15
electronically signed data construct
structured data set containing the header, payload, signature and optional data fields

Note 1 to entry: The payload type and issuer identity are included in the header.

Note 2 to entry: An ESDC is a machine-readable code.

Note 3 to entry: An ESDC can contain non-signed data elements.

[SOURCE: CEN EN 18246:2026, 3.1[61] , modified]

7.16
validation
confirmation of the plausibility for a specific intended use or application through the provision of objective evidence that specified requirements have been met

Note 1 to entry: Compared with the source: numbers referencing to related sections and the accompanying note have been removed from the definition.

[SOURCE: ISO/AWI 25534-1:2026, 3.6, modified — ISO/IEC 17000:2020, 6.5, modified]

7.17
verification
confirmation of the truthfulness through the provision of objective evidence that specified requirements have been fulfilled

Note 1 to entry: Compared with the source: numbers referencing to related sections and the accompanying note have been removed from the definition.

[SOURCE: ISO/AWI 25534-1:2026, 3.7, modified — ISO/IEC 17000:2020, 6.6, modified]

8 Terms related to semantics and profiles

8.1
interoperability
ability of two or more systems or applications to exchange information and to mutually use the information that has been exchanged

Note 1 to entry: Regional variant: the definition recorded from the merged concept pair was: capability of two or more entities to exchange items in accordance with a set of rules and mechanisms implemented by an interface in each entity, in order to perform their specified tasks

[SOURCE: ISO/IEC 22123-1:2023, 3.6.1, modified — CEN EN 18223:2026, 3.1, similar, ISO 18435-1:2009, 3.12, similar, CEN EN 18219:2026, 3.1.7]

8.2
semantic interoperability
ability for data shared by systems to be understood at the level of fully defined domain concepts

[SOURCE: ISO 18308:2011, 3.45, modified — CEN EN 18223:2026, 3.2]

8.3
technical interoperability
ability that covers the applications and infrastructures linking systems and services

[SOURCE: CEN EN 18223:2026, 3.3[59]]

8.4
application programming interface
set of methods (*method* (8.6)) provided by an application component for use by other application components

[SOURCE: CEN EN 18222:2026, 3.1[58]]

8.5
representational state transfer application programming interface
service that allows for interaction with resources via a stateless, client-server architecture, typically using standard HTTP calls like GET, POST, PUT, and DELETE to perform methods (*method* (8.6)) by an application component on these resources

[SOURCE: CEN EN 18222:2026, 3.1.1[58]]

8.6
method
particular way used to perform a specific action on a resource

EXAMPLE read a DPP

[SOURCE: CEN EN 18222:2026, 3.3[58]]

8.7
parameter
data provided by a client, that is needed to perform a method (*method* (8.6))

EXAMPLE DPP ID for read DPP

[SOURCE: CEN EN 18222:2026, 3.4[58]]

8.8
result
return value after the execution of a method (*method* (8.6))

EXAMPLE the DPP

[SOURCE: CEN EN 18222:2026, 3.5[58]]

8.9
data exchange
storing, accessing, transferring, and archiving of data

[SOURCE: ISO 15531-43, 3.1.5, modified — CEN EN 18216:2026, 3.2]

8.10
application
software functional element specific to the solution of a problem in industrial-process measurement and control

Note 1 to entry: An application can be distributed among resources and may communicate with other applications.

[SOURCE: IEC TR 62390:2005, 3.1.2, modified — IDTA 01001-3.2:2026, 1.1.2]

8.11
asset
entity owned by or under the custodial duties of an organization, which has either a perceived or actual value to the organization

EXAMPLE 1 Examples for physical entities are equipment, raw material, parts components and pieces, supplies, consumables, physical products, and waste.

EXAMPLE 2 Software is an example of a digital asset.

EXAMPLE 3 A software license is an example of an intangible asset.

Note 1 to entry: An asset can be single entity, a collection of entities, an assembly of entities or a composition of entities.

[SOURCE: IEC 63278-1:2023]

8.12
attribute
data element of a property, a relation, or a class in information technology

[SOURCE: IEC 63278-1:2023]

8.13
Asset Administration Shell
standardized digital representation of an asset

Note 1 to entry: Asset Administration Shell and Administration Shell are used synonymously.

[SOURCE: IEC 63278-1:2023]

8.14
class
description of a set of objects that share the same attributes, operations, methods, relationships, and semantics

[SOURCE: IEC TR 62390:2005, 3.1.4, modified — IDTA 01001-3.2:2026, 1.1.6]

8.15
capability
implementation-independent potential of a component to achieve an effect within a domain

Note 1 to entry: Capabilities can be orchestrated and structured hierarchically.

Note 2 to entry: Capabilities can be made executable via services.

Note 3 to entry: The impact manifests itself in a measurable effect within the physical world.

[SOURCE: Glossar Industrie 4.0[47]]

8.16
coded value
value that can be looked up in a dictionary and can be translated

[SOURCE: ECLASS[45]]

8.17
component
product used as a constituent in an assembled product, system, or plant

[SOURCE: IEC 63278-1:2023, 3.6, modified — IDTA 01001-3.2:2026, 1.1.9]

8.18
concept
unit of knowledge created by a unique combination of characteristics

[SOURCE: IEC 63278-1:2023]

8.19
digital representation
information and services representing an entity from a given viewpoint

EXAMPLE 1 Examples of information are properties (e.g. maximum temperature), actual parameters (e.g. actual velocity), events (e.g. notification of status change), schematics (electrical), and visualization information (2D and 3D drawings).

EXAMPLE 2 Examples of services are asset services (for example providing the history of the configuration data or providing the actual velocity) and asset related services (for example providing a simulation).

EXAMPLE 3 Examples of viewpoints are mechanical, electrical, or commercial characteristics.

[SOURCE: IEC 63278-1:2023]

8.20
digital twin
digital representation, sufficient to meet the requirements of a set of use cases

Note 1 to entry: In this context, the entity in the definition of digital representation is typically an asset.

[SOURCE: IDTA 01001-3.2:2026, 1.1.12[50]]

8.21
explicit value
commonly used concept, like numbers (e.g. 109, 25) which do not need lookup in dictionaries

[SOURCE: ECLASS[45]]

8.22
instance
concrete, clearly identifiable component of a certain type

Note 1 to entry: An individual entity of a type, for example a device, is obtained by defining specific property values.

Note 2 to entry: In an object-oriented view, an instance denotes an object of a class (of a type).

[SOURCE: IEC 62890:2016]

8.23
instance asset
specific asset that is uniquely identifiable

EXAMPLE Examples of instance assets are material, a product, a part, a device, a machine, software, a control system, or a production system.

[SOURCE: IEC 63278-1:2023]

8.24
operation
executable realization of a function

Note 1 to entry: The term method is synonymous to operation.

Note 2 to entry: An operation has a name and a list of parameters [ISO 19119:2005, 4.1.3].

[SOURCE: Glossar Industrie 4.0[47]]

8.25
property
defined characteristic suitable for the description and differentiation of products or components

Note 1 to entry: The concept of type and instance applies to properties.

Note 2 to entry: This definition applies to properties as described in IEC 61360/ ISO 13584-42.

Note 3 to entry: The property types are defined in dictionaries (like IEC component data dictionary or ECLASS), they do not have a value. The property type is also called data element type in some standards.

Note 4 to entry: The property instances have a value and are provided by the manufacturers. A property instance is also called property-value pair in certain standards.

Note 5 to entry: Properties include nominal value, actual value, runtime variables, measurement values, etc.

Note 6 to entry: A property describes one characteristic of a given object.

Note 7 to entry: The specification of a property can include predefined choices of values.

[SOURCE: Glossar Industrie 4.0[47]]

8.26
qualifier
well-defined element associated with a property instance or SubmodelElement, restricting the value statement to a certain period of time or use case

Note 1 to entry: Qualifiers can have associated values.

[SOURCE: IEC 62569-1:2017]

8.27
service
distinct part of the functionality that is provided by an entity through interfaces

[SOURCE: IEC 63278-1:2023, Clause 741-01-28, modified — IDTA 01001-3.2:2026, 1.1.20]

8.28
Submodel
representation of an aspect of an asset

[SOURCE: IEC 63278-1:2023]

8.29
SubmodelElement
element of a Submodel

[SOURCE: IEC 63278-1:2023]

8.30
Submodel template
template for the representation of an aspect of an asset

[SOURCE: IEC 63278-1:2023]

8.31
Submodel template element
element of a Submodel template

[SOURCE: IEC 63278-1:2023]

8.32
system
interacting, interrelated, or interdependent elements forming a complex whole

[SOURCE: IEC 63278-1:2023, 3.2.123, modified — IDTA 01001-3.2:2026, 1.1.25]

8.33
template
specification of the common features of an object in sufficient detail that such object can be instantiated using it

Note 1 to entry: Object can be anything that has a type.

[SOURCE: ISO/IEC 10746-2]

8.34
type
hardware or software element which specifies the common attributes shared by all instances of the type

[SOURCE: IEC TR 62390:2005, 3.1.25, modified — IDTA 01001-3.2:2026, 1.1.27]

8.35
type asset
(abstract) representation of a set of instance assets with common characteristics and features

EXAMPLE Examples of type assets are type of material, a product type, a type of a part, a device type, a machine type, a type of software, a type of control system, a type of production system.

Note 1 to entry: The set of instance assets may exist or may not exist.

[SOURCE: IEC 63278-1:2023]

8.36
variable
software entity that may take different values, one at a time

[SOURCE: IEC 61499-1:2012]

9 Terms related to measurement and material flow

9.1
mass balance model
chain of custody model in which materials or products with a set of specified characteristics are mixed according to defined criteria with materials or products without that set of characteristics

Note 1 to entry: The proportion of the input with specified characteristics can only match the initial proportions on average and will typically vary across different outputs.

[SOURCE: ISO 22095:2020, 3.3.4, modified — CEN CWA 18291:2025, 3.28]

10 Terms related to the framework: profiles, lenses and jurisdiction

10.1
profile
registered, versioned object that binds a set of data points, transforms, cryptographic suites, trust requirements, access policies, presentation requirements and a trigger predicate to the neutral core passport, constraining and transforming, but never redefining, the single definition of each data point

Note 1 to entry: Composition is three-axial: jurisdiction, sector, characteristic.

Note 2 to entry: Each profile is a projection function, view(twinState, profile, actor), whose projection may transform and not merely select.

Note 3 to entry: Profiles constrain and transform; inter-profile disagreement is expected and auditable, while intra-class contradiction is an integrity incident (single definition, many constraints).

10.2
lens
profile conceived as a calibrated measuring instrument placed on the digital twin, which selects and transforms without altering the observed scene

Note 1 to entry: The mount, the profile interface on the neutral core, is the standardized element: any registered lens by any maker attaches to it.

Note 2 to entry: A lens publishes its calibration (version and cited rules), its uncertainty and its decision rules; it is subject to pattern approval at registration and to periodic re-verification through conformance runs.

Note 3 to entry: Verification path-finds from any render through the lens and its fact chain to an anchor in the verifier's trust bundle.

10.3
lens mount
standardized interface of the neutral core through which any registered profile is attached to and operates on the record of the subject, such that profiles from any registering authority compose without proprietary binding

Note 1 to entry: The mount, not any single profile technology, is the interoperation point; standardizing it precludes lock-in to a proprietary profile mechanism.

10.4
jurisdiction profile
profile that binds the data points, transforms, cryptographic suites, trust requirements, access rules, presentation and carrier requirements enacted by the law of a given jurisdiction

Note 1 to entry: Conflicting regional rules are registered as distinct data elements harmonized to a common concept; they are not merged.

10.5
sector overlay
profile that composes with other profiles to add the data points, transforms and rules applicable to an industry sector, such as electronics, textiles, batteries, construction, tyres or packaging

10.6
applicable law
totality of legal rules that apply to a subject placed on a market, represented in the passport system by the corresponding jurisdiction profile and its dated applicability bindings

10.7
as-of query
query against the event-sourced record of a subject that reconstructs the state that was legally valid at a specified past time

Note 1 to entry: As-of queries are answered from notarized snapshots, log anchoring and recorded profile-version bindings, for example to establish whether a product was compliant when sold.

10.8
trust list
signed, registry-deposited list of the trust anchors accredited to act under a given profile or jurisdiction, served with status and revocation information

Note 1 to entry: Jurisdictional trust lists relate to one another through a multi-witnessed master list.

11 Terms related to the framework: projection, composition and service tiers

11.1
projection
deterministic, registered and legally cited view of the testimonial record of a governed digital twin, computed under a profile from the event-sourced change log

Note 1 to entry: The passport is the legal materialized view of the twin; the twin is the source of truth.

Note 2 to entry: Notarized snapshots support as-of queries, which answer what was legally true at a given time.

Note 3 to entry: Any conforming twin can emit the view; the projection is defined representation-neutrally.

11.2
materialized view
rendered, citable result of a projection, materialized at a point in time for serving and verification

Note 1 to entry: The digital product passport is the legal materialized view of the digital twin.

11.3
child passport
passport of a component or sub-product that is independently placed on the market or independently regulated, referenced by identity from the passport of the composite product

Note 1 to entry: Children join by identity reference, never by data copying; roll-up views are computed from the references.

11.4
containment event
typed event recording that one passport is physically or structurally contained in another, retained for genealogy and recall traversal and never transferring authority

11.5
genealogy
complete directed ancestry of a passport, traced through transformation and containment events from raw-material and component passports to the present subject

11.6
as-of state
state of a subject reconstructed from its event-sourced record at a specified past time, anchored by a notarized snapshot so that queries can answer what was legally true at that time

Note 1 to entry: The two authoritative query modes are the live view (freshness-labelled) and the notarized as-of snapshot (log-anchored); legal purpose selects the mode.

Note 2 to entry: Transformation events hash-link their inputs by as-of state hash, and profile-version bindings recorded as lifecycle events make manifest history as-of-reconstructable, answering questions such as whether a product was compliant when sold.

Note 3 to entry: Mirrors and caches are always as-of stamped; the projection is computed from the record, so as-of queries are reproducible.

11.7
offline minimum
subset of passport data and verification material specified as sufficient for offline verification of identity and status from the data carrier alone

Note 1 to entry: The Tier-A payload is the carrier-embedded realization of the offline minimum.

11.8
degradation ladder
ordered set of service tiers by which availability and verification degrade explicitly, from the served full passport through the carrier-embedded minimum to the archival record

Note 1 to entry: Stale or offline data is always marked as such, never silently passed as current.

11.9
Tier-A payload
carrier-embedded minimum viable passport, comprising the product identifier, the resolver URI, the economic operator identifier, the status, the critical safety or recall flag, the validity and multi-suite compressed signatures, sized to the data carrier and verifiable offline against pre-cached trust anchors

Note 1 to entry: The three-tier degradation ladder comprises Tier A (carrier-embedded minimum), Tier B (served full passport) and Tier C (archival).

Note 2 to entry: Stale or offline data degrades explicitly through freshness verdicts and never passes silently.

11.10
Tier-B payload
full passport content, served through resolution and hosting services in accordance with the applicable profile

11.11
Tier-C archive
notarized archival form of the passport record, retained for persistence beyond the operational lifetime of any single provider

12 Terms related to the framework: provenance, identity lattice and edge segments

12.1
provenance class
declared category of a fact according to how it entered the record: declared at type level, measured at instance level under a stated method and uncertainty, or computed by a stated transform

12.2
precedence rule
rule declared by a profile stating which provenance class prevails when facts about the same subject disagree

Note 1 to entry: The default precedence is instance-measured over type-declared; a genuine same-class contradiction is an integrity incident, never a silent merge.

12.3
roll-up attestation
signed attestation giving an aggregate over a committed traversal set of child passports, verifiable without fetching every member of the set

12.4
traversal set
committed, hash-identified set of passport versions over which a roll-up attestation or a graph traversal is computed

12.5
edge segment
part of the event-sourced record of a subject that originates on the subject's own device and is published through device commitments

Note 1 to entry: The device is a tier of the passport system, not a client; it commits now and reveals later, but never contradicts.

12.6
device commitment
signature by a device-bound key over the hash of a prefix of the device's local append-only event log, published so that later disclosures are inclusion-consistent with past commitments

12.7
configuration vector
exact combination of registered component options that, together with a model, constitutes a configured product instance

12.8
type version
versioned state of a type passport capturing hardware revisions and type-declared facts, referenced exactly by the instance passports derived from it

12.9
derived type
new type resulting from a regulated modification of an existing type, recorded as a derivation event in the identity lattice

12.10
dormant identifier
identifier recorded at the finest available granularity for an object class for which no passport regime yet exists, held as issuance evidence for adoption by a future regime

Note 1 to entry: When a future regime issues passports for the object class it adopts the recorded dormant identifiers rather than minting new ones, so that the installed base connects retroactively through its build records.

Note 2 to entry: Inputs are recorded even when no passport exists behind them yet; minting fresh identifiers would orphan the installed base.

12.11
dormant-to-live adoption
act of a future passport regime of issuing passports by adopting the identifiers already recorded as dormant for the object class, rather than minting new ones

Note 1 to entry: Adoption connects the installed base retroactively through its build records; minting fresh identifiers would orphan it.

13 Terms related to the framework: stamps and the transformation algebra

13.1
stamp
lens-scoped attestation consisting of an attester signature over the subject, the subject state commitment, the lens identity and version, the mode (live or snapshot), a verdict with coverage report, and log-anchored time

Note 1 to entry: Stamps accrete append-only and are valid under the stamper's lens regardless of the issuer's.

Note 2 to entry: Jurisdiction stamps on a foreign-issued passport correspond to visas; entry and exit stamps correspond to verifier custody events.

Note 3 to entry: Stamps may carry quantity context, inherited by downstream transformations.

13.2
lens-scoped attestation
attestation whose validity is defined by, and only by, the profile under which it was issued

Note 1 to entry: A stamp is the signed form of a lens-scoped attestation.

13.3
derived passport
passport issued as the output of a transformation event by an actor accredited for the regulated claims it contains, hash-linking its input passports

Note 1 to entry: Only accredited actors may issue derived passports for regulated claims.

Note 2 to entry: Children of a split carry a derived-from reference and quantity carve-outs, with the sum of children not exceeding the parent (remainder semantics).

13.4
transformation event
typed event in the provenance directed acyclic graph that records the split of one passport into many, or the combination or transformation of many passports into one

Note 1 to entry: The output issuance event carries input references of passport, quantity and as-of state hash, so the new passport hash-links its sources.

Note 2 to entry: Quantities resulting from a transformation are new measured facts with provenance "computed by transformation event", never copies; mass balance is auditable as input minus output equals loss.

Note 3 to entry: Inputs transition to consumed or transformed status: historical, still verifiable, still needed for as-of queries.

13.5
inputReference
record in the issuance event of a derived passport naming an input passport, the quantity consumed and the as-of state hash under which it was taken

Note 1 to entry: The new passport hash-links its sources, so that the provenance graph is reconstructable without trusting the issuing actor alone.

13.6
quantity carve-out
recorded allocation of input quantity to an output of a split transformation, such that the sum of the child quantities does not exceed the parent

13.7
mass balance accounting
bookkeeping over transformation events in which inputs and outputs of a specified characteristic are reconciled per period, with losses and taint carried explicitly

Note 1 to entry: Mass balance accounting applies a mass balance model (3.7.1) to a set of transformation events.

13.8
consumed passport
input passport that has reached the consumed or transformed state in a transformation event, retaining historical verifiability for as-of queries

13.9
blind provenance
attestation of input provenance issued by a notified or accredited body that withholds the raw input identities behind its own attestation, used where commercial confidentiality demands it

13.10
jurisdiction stamp
stamp issued under a jurisdiction profile on a passport issued under another jurisdiction, recording that the stamping jurisdiction's requirements were assessed

Note 1 to entry: The colloquial designation of a jurisdiction stamp on a foreign-issued passport is “visa”; entry and exit stamps are the verifier's custody events.

14 Terms related to the framework: characteristic profiles and orphan objects

14.1
characteristic profile
profile that attaches by predicate on digital twin facts, such as age, material content, heritage status or market status, rather than by law-of-place or industry

Note 1 to entry: Characteristic profiles split into regulatory-triggered profiles (for example CITES species content, cultural-goods import controls, art-market due diligence, conflict-minerals origin) and voluntary value profiles (for example provenance and grading for antiques and collectibles).

Note 2 to entry: Predicate triggers include time predicates: an object becoming older than a given age is a clock-fired applicability event handled by the dated-binding machinery.

14.2
trigger predicate
declared component of a profile stating the conditions under which the profile applies, evaluated against the facts of the digital twin rather than against who declares them

Note 1 to entry: Characteristic profiles attach by predicate on twin facts, namely age, material content, heritage status, market status, rather than by law-of-place or industry.

Note 2 to entry: Predicate triggers include time predicates: an object becoming older than a given age is a clock-fired applicability event, handled by the dated-binding machinery so that applicability attaches without human declaration.

Note 3 to entry: The expression “characteristic profile trigger” denotes this predicate as used by a characteristic profile and is folded into this concept rather than recorded separately.

14.3
commissioning
issuance act by which a qualified attestor or registrar establishes the identity of an object that predates or lies outside a manufacturer-issued regime, recording the first sighting in the event log

14.4
qualified attestor
actor accredited under a profile's trust list to commission identities or to attest attributes of orphan objects, such as an auction house, authentication board or registrar

14.5
attribution
expert statement assigning an object to a creator, workshop or period, recorded as a competing fact with provenance rather than merged with other attributions

14.6
provenance gap
explicitly recorded interval in a subject's custody or provenance chain for which no evidence exists

Note 1 to entry: A provenance gap is data, not an error; it is honestly representable in the event-sourced record.

14.7
condition report
dated, signed, profile-scoped assessment of the physical condition of an object

14.8
restitution flag
authority-attached marker on a passport recording a claim over title, such as looting, dispossession or colonial taking, propagated as a graph event without rewriting history

15 Terms related to the framework: capability classes and digital twins

15.1
capability class
classification of a subject by its ability to participate in its own record: silent, passive-auth, logged-contact or connected

Note 1 to entry: A profile's freshness and verification requirements shall be satisfiable by the capability class of the subject; demanding live freshness from a silent subject is an unsatisfiable profile.

15.2
silent subject
subject with no connectivity, keys or self-reporting, whose record is reconstructed entirely from testimonies about it

15.3
passive-auth subject
subject carrying an authentication element, such as an NFC chip, physically unclonable function or identity link, that attests identity without recording events

15.4
logged-contact subject
subject that records events locally and discloses them on physical contact, without communications capability

15.5
connected subject
subject that self-reports under device keys and participates with full edge segments

15.6
passive twin
digital twin reconstructed from point-in-time attestations about the object, sampled when the object visits an attestor, in which truth is episodic and staleness between attestations is unbounded and undetectable

Note 1 to entry: Corresponds to capability classes S0 (silent, testimony-only) and S1 (passive authentication).

Note 2 to entry: Trust rides entirely on attestor authority; the twin is testimonial by default and sensorial by exception.

15.7
active twin
digital twin in which the object self-reports under device keys, making truth near-continuous, freshness bounded and detectable, and shifting the binding question from who saw the object to who attests the sensor

Note 1 to entry: Corresponds to capability classes S2 (logged contact) and S3 (connected).

Note 2 to entry: Measurement validity becomes load-bearing: sensor calibration, uncertainty of self-reported values and registered units; health status is a derived verdict.

Note 3 to entry: Profiles declare, per data element, the required truth mode, namely attest-sampled, self-committed, or both with precedence.

15.8
truth mode
declared per-data-element attribute of a profile stating which truth source the element requires, namely attest-sampled, self-committed, or both with a precedence rule

Note 1 to entry: The passive twin (concept dpp-0132) supplies attest-sampled truth and the active twin (concept dpp-0133) supplies self-committed truth; both paradigms coexist permanently in the framework.

Note 2 to entry: Precedence between a calibrated device measurement and an attestor observation is itself a metrological hierarchy question.

Note 3 to entry: Freshness and verification requirements must be satisfiable by the subject's capability class: demanding live freshness from an S0 (silent) product is an unsatisfiable profile.

15.9
service-center model
arrangement in which a subject's state is sampled and attested only when the subject visits an attestor, yielding episodic truth and unbounded, undetectable staleness between attestations

15.10
self-testimony
testimony class consisting of events asserted by the subject's own device under device-bound keys

15.11
continuous conformity monitoring
conformity mode in which requirements are re-judged against the live state of the twin, rather than only at point-in-time inspection

15.12
state of health
derived verdict on the condition of a subject, computed by a stated model from measured variables

Note 1 to entry: The provenance of the computing model is itself governed information; health status is not a raw fact.

15.13
sensor attestation
attestation binding a measured value to a registered sensor identity, its calibration and its measurement uncertainty

16 Terms related to the framework: passport relationship algebra

16.1
association
typed, navigational passport relationship, such as compatibility, supply or same-maker, carrying no lifecycle impact

16.2
installation
typed structural relationship, and its event class, between a parent passport and an installed child component, recorded with method, slot identity, pairing, alterations, performer and time, which makes an installed component a different legal object from the standalone one

Note 1 to entry: Schema: install(child, parent, method, slotId, pairing (none, firmware, key or module), alterations, performedBy, at).

Note 2 to entry: Bidirectional parent knowledge is mandatory: the parent's manifest lists children and the child's event log records its installation interval (parent, slot, method), for recall routing, scoped service access, ownership transfer and end-of-life routing.

Note 3 to entry: Lifecycle coupling: warranty re-scope and service economics ride the parent, and the part is no longer sellable as new; custody transfer (concept dpp-0134) remains orthogonal, since ownership changes while composition stays fixed.

Note 4 to entry: The recoverability of the binding (concept dpp-0143) determines identity continuity: disassembly that restores a marketable object with identity continuity is installation, while dissolving identity is a transformation event (concept dpp-0129). Part replacement is an uninstall followed by an install, and regulated children keep their own passports through it.

16.3
membership
temporal relationship between a passport and a group node, such as a shipment, consignment, kit, fleet or recall set

16.4
group node
queryable grouping of passports that receives a passport of its own only when the group is itself placed on the market

16.5
binding strength
declared permanence of an installation binding, ranging from reversible fastening to destructive integration

16.6
slot identity
parent-scoped identifier of an installation position by which an installed child is re-identified within its parent

16.7
pairing
recorded binding of a child to its parent context, by firmware, key or module, that is re-established when a component is replaced

16.8
absorbed component
input whose identity dissolves into its parent and is therefore modelled as a transformation input rather than a live child, recorded at the finest available granularity as dormant identifiers

16.9
recoverability
declared property of an installation binding (concept dpp-0142) stating whether and how the installed child can resume an independent existence, taking one of the values restorable, harvestable, destructive or absorbing

Note 1 to entry: Restorable: the child resumes standalone life; harvestable: recovered but altered, continuing with harvested status and parent history (concept dpp-0144); destructive: the child ceases and flows to a material passport through a transformation event (concept dpp-0129); absorbing: identity dissolves into the parent.

Note 2 to entry: Absorption is regime-temporal rather than permanent: absorbed inputs are recorded at the finest available granularity as dormant identifiers (concept dpp-0135), issuable by adoption when a future regime requires passports for the object class.

Note 3 to entry: Carried in the EXPRESS core as part of the PassportLink binding (method, recoverability).

16.10
harvested part
component recovered from an installation (concept dpp-0142) in the harvestable state of the recoverability spectrum (concept dpp-0143), which continues to carry its own identity, harvested status and the parent history accumulated during installation

Note 1 to entry: Installation history is value-relevant provenance: a harvested battery sold standalone carries its service interval and removal reason, and the used-parts market runs on this.

Note 2 to entry: A harvested part is second-hand-once-removed: no longer sellable as new, with warranty scope and service economics that rode the parent during installation.

16.11
custody transfer
signed ceremony recording the transfer of responsibility for an object between custodians, appended to the event chain by the custodian and the counterparty, which extends and never overwrites the trail across successive owners

Note 1 to entry: Custody links are social and control edges, orthogonal to structural composition: ownership changes while composition stays fixed.

16.12
custodian
actor that currently holds the live event segment of a subject's record and is authorized to append custody-relevant events to it

17 Terms related to the framework: visibility and enumeration resistance

17.1
blind edge
edge on a typed passport relationship whose existence and endpoints are concealed from all observers, so that a child-to-parent reference does not reveal where a thing is, such as a household or a site

Note 1 to entry: Edge visibility classes, namely public, restricted (role-qualified), blind and escrowed, are declared on every relationship; the installation schema (concept dpp-0142) carries visibility as an edge class with escrow arrangement (concept dpp-0137) and audiences.

Note 2 to entry: Blind is the default for consumer installations, because even the object-to-household linkage is personal data requiring a GDPR or PIPL legal basis per profile.

Note 3 to entry: Rationale: a traversable graph with entitled upstream observers is surveillance infrastructure; post-sale visibility into instance state is owner-push or authority-based, never maker-pull.

17.2
escrowed disclosure
edge visibility arrangement in which information concealed by a blind edge (concept dpp-0136) is deposited with a threshold trustee and released only through a disclosure ceremony, namely court, regulator or consent

Note 1 to entry: The installation schema records escrow as none or trustee, with declared audiences.

Note 2 to entry: An escrow envelope under a threshold trustee may optionally accompany a proof of binding (concept dpp-0138), preserving verifiability while full disclosure remains ceremonial.

17.3
proof of binding
salted cryptographic commitment to the parent, recorded in the child's event log, by which a child satisfies the bidirectional parent-knowledge requirement without disclosing the identity of the parent

Note 1 to entry: Proof of binding is distinct from knowledge of parent: verifiable modes establish that the child is validly installed in some parent under a given profile (warranty, compliance) without revealing which parent; full disclosure occurs only by ceremony.

Note 2 to entry: An optional escrow envelope under a threshold trustee (concept dpp-0137) accompanies the commitment where future disclosure must remain possible.

17.4
enumeration resistance
system property by which the passport system is verifiable without being browsable: resolution is by identity only, traversal requires per-edge rights, and no party, including any registry, resolver network or log operator, can enumerate the installed base of anything

Note 1 to entry: Transparency logs anchor commitments (hashes), never facts, so log operators cannot correlate edges; accumulator and blind-membership proofs give valid-standing checks without revealing which or where.

Note 2 to entry: Stated as design invariant I12, together with blind edges by default where households or sites are revealed (concept dpp-0136), proof of binding distinct from knowledge of parent (concept dpp-0138), and predicate-based recall (concept dpp-0140).

17.5
predicate-based recall
recall scheme in which the recall set is never enumerated centrally: the recall predicate is published, and each custodian evaluates it locally against their own holdings, including dark and defence holdings under national procedure, and acts

Note 1 to entry: The manufacturer receives aggregates or nothing; there is no traversal-push enumeration of the installed base.

Note 2 to entry: One of the mechanisms by which enumeration resistance (concept dpp-0139) is maintained: recall propagation still notifies custodians transitively, but through local predicate evaluation rather than central construction of the recall set.

17.6
owner-push disclosure
disclosure of instance state initiated by the owner, such as a warranty registration, as the default manufacturer-visibility channel

Note 1 to entry: A manufacturer-initiated pull of instance state is not a channel of the framework; authority-based compulsion of traversal is a separate, legal-basis-governed mechanism.

17.7
dark identity
identity that is resolvable only within a restricted resolution domain, such as a national domain, and never enters the global resolver

17.8
confidential profile
registered profile whose register item is non-public and which carries restricted resolution, edge-visibility and traversal fields, used for defence and sovereignty configurations

18 Terms related to the framework: trust semantics, degradation and federation

18.1
trust marker
graded indicator attached to every element or event stating its trust provenance: unsigned, self-declared, third-party attested, multi-signed or log-anchored

Note 1 to entry: Trust is graded, not mandated; profiles set minimum signing per data class.

18.2
master list
globally multi-witnessed list of trust lists, co-signed by a quorum of independent logs, by which verifiers cross-recognize jurisdictional trust authorities

18.3
revocation window
explicit start and end interval carried by a distrust declaration, within which affected credentials are void and outside which they are re-validated

Note 1 to entry: The reason for revocation determines retroactivity: prospective reasons preserve earlier as-of verifications; misissuance voids validity from the beginning.

18.4
three verification readings
three legally distinct readings of a verification result, namely evidentiary, current-state and cryptographic, each of which a verdict states explicitly

Note 1 to entry: The evidentiary reading asks what a diligent verifier could know at the time; the current-state reading voids fraud from the beginning; the cryptographic reading covers signature and chain validity alone.

18.5
freshness verdict
explicit, displayable assessment of how current a served state is relative to its declared freshness bound, degrading visibly when data is stale or offline

18.6
coverage report
statement accompanying a verdict enumerating which elements, trust anchors and registry items the verification covered

18.7
discovery registry
federated registry of record for passport services and semantics, in which profile shapes, protocol bindings, verification mechanisms and service endpoints are versioned, signed, log-anchored items discoverable by any conforming client

Note 1 to entry: The registry holds descriptors of services and shapes, never records of things.

18.8
service descriptor
signed registry item describing a service by operator identity and credential, service class, endpoints, protocol binding, jurisdiction, residency class, status and succession pointer

18.9
protocol binding
registered specification of a wire grammar, media types, version and conformance suite by which an interaction with a service or a device is conducted

18.10
verification mechanism bundle
assembled set of registered items naming the cryptographic suite, trust framework, trust-list endpoints, revocation mechanism and acceptance-policy template under which a client verifies

18.11
listing gate
registered predicate evaluated by a marketplace over passport status and standing without holding passport data, admitting or refusing listings

18.12
operator credential
scoped, threshold-issued credential by which an operator signs registry items and service attestations

Note 1 to entry: Appends claiming a role outside the credential's scope degrade the trust marker; they do not pass silently.

18.13
onboarding ceremony
threshold admission ceremony by which an operator joins the service federation, or a trust authority joins the master list, accepting continuity and succession obligations

18.14
read federation
federation tier in which any party consumes the registry and trust bundles without an admission ceremony

18.15
service federation
federation tier in which operators register services under signed descriptors and accept continuity and succession obligations

18.16
trust federation
federation tier in which trust authorities join the master list by threshold admission and jurisdictional profiles cross-recognize one another

18.17
seed bundle
bootstrap set of registry endpoints and trust anchors with which a conforming client first initializes, resolving the circular dependence between registry discovery and trust

18.18
registry pinning
binding of a passport manifest or a verdict to exact registry items by item, version and hash, so that later registry changes never alter what a past issuance must satisfy

Annex A
(normative)

Sources and harmonization matrix

A.1 Corpus and method

The terms and definitions clauses of the following source documents were harvested in full and verified concept by concept. Table A.1 records, per source document, the number of concepts in this document that cite it and the clauses cited. The harvest comprised 141 term entries (82 from the EN 182xx series, 30 from CWA 18291:2025 and 29 from IDTA 01001-3.2:2026); ISO/AWI 25534-1 contributes the conflicting definition of the head term recorded in 3.1, together with the terms newly introduced in 3.4 (life cycle, circular economy, product circularity, traceability, value chain) and 3.5 (validation, verification) that were not previously represented in the harvested corpus. Concepts carried forward from the harvest carry source citations in their entries; lineage citations (standards from which a source document itself adopted a definition) are recorded in the source blocks of the entries.

Table A.1 — Source documents and concepts citing them

Source document	Concepts	Principal clauses cited
EN 18219:2026, *Unique identifiers*	24	Clause 3
EN 18220:2026, *Data carriers*	19	Clause 3
CWA 18291:2025, textile supply chains	29	Clause 3
IDTA 01001-3.2:2026, AAS metamodel	29	Clause 1.1
EN 18239:2026, *Access rights management, information system security, and business confidentiality*	9	Clause 3
EN 18216:2026, *Data exchange protocols*	8	Clause 3
EN 18222:2026, *Application Programming Interfaces (APIs)*	7	Clause 3
EN 18221:2026, *Data storage, archiving, and data persistence*	5	Clause 3
EN 18223:2026, *System interoperability*	4	Clause 3
EN 18246:2026, *Data authentication, reliability and integrity*	3	Clause 3
ISO/AWI 25534-1 (working draft 2026-09-05)	8	Clauses 3.1 to 3.8
IEC 63278-1:2023 (lineage citations, incl. compound sources)	14	Clause 3 and Annex

Lineage standards cited by entries of this document include IEC 63278-1:2023, ISO 59004:2024, ISO 5157:2023, ISO 29404:2015, ISO 22095:2020, ISO 59040, ISO/IEC 19762, ISO/IEC 17000:2020, ISO/IEC 22123-1:2023, ISO 18435-1:2009, ISO 18308:2011, IEC 61360-1:2016, ISO 22274:2013, ISO/IEC Guide 77-2:2008, IEC 61666:2010+AMD1:2021 CSV, IEC 62569-1:2017, IEC 62890:2016, IEC 61499-1, IEC 60050-741:2020, ISO/IEC 15459-1, ISO/IEC 18975:2024, ISO/IEC 20248:2022, ISO 9000:2015, ISO 14040:2006, ISO 80000-3, ISO 8601-1:2019, UNECE Recommendation No. 46 and the UN Transparency Protocol; the complete list is given in the Bibliography.

A.2 Recorded conflicts and their harmonization

The following conflicts were recorded during verification and harmonization.

— **Head term.** EN 18223:2026, EN 18239:2026, EN 18216:2026, EN 18221:2026, EN 18222:2026 and EN 18246:2026 define the digital product passport as a “digital record of product characteristics throughout its life cycle”; EN 18219:2026 and EN 18220:2026 extend the same definition by accessibility “via electronic means through a data carrier”; ISO/AWI 25534-1 defines it as “a concept or technology which enables sharing of product related information”. The ISO/AWI 25534-1, 3.1 formulation is the preferred definition of the entry at 3.1.1; the EN-family record-side variants are retained as source citations with `status: not_equal` and per-source modification notes recording the substantive variance; the EN 18219 / EN 18220 extension is recorded with its carrier-access qualifier. Resolution of the conflict is a decision for ISO/IEC JTC 5, for which this document is the input.

— **Unique identifier.** EN 18219:2026, 3.1.24 and EN 18220:2026, 3.18 define the unique identifier as a "string of characters that is required to be unique among all identifiers used for all objects for a specific purpose"; CWA 18291:2025, 3.3 and the underlying ISO 29404:2015, 3.26 (and, behind that, the ISO/IEC 15459 family) define it as an "identifier which is guaranteed to be unique among all identifiers used for those objects and for a specific purpose". The ISO 29404:2015, 3.26 formulation is the preferred definition of the entry at 3.2.2; the EN-family formulation is recorded as a regional variant with `status: modified` and the substantive variance (`identifier` → `string of characters`; `those objects` → `all objects`; Note 1 to entry replaced) named in the source block.

— **Economic operator.** EN 18219:2026, 3.1.5 and EN 18220:2026, 3.7 define the economic operator with the qualifier "that places the product on the market"; EN 18221:2026, 3.1 defines it without that qualifier. The EN 18221:2026, 3.1 formulation (the broader definition) is the preferred definition of the entry at 3.3.1; the narrower EN 18219 / EN 18220 formulation is recorded as a regional variant with `status: similar` and the qualifier named in the source block.

— **Interoperability.** EN 18219:2026, 3.1.7 (with ISO/IEC 22123-1:2023, 3.6.1 lineage) defines interoperability as "the ability of two or more systems or applications to exchange information and to mutually use the information that has been exchanged"; EN 18223:2026, 3.1 (with ISO 18435-1:2009, 3.12 lineage) defines it as "the capability of two or more entities to exchange items in accordance with a set of rules and mechanisms implemented by an interface in each entity, in order to perform their specified tasks". The ISO/IEC 22123-1:2023, 3.6.1 formulation is the preferred definition of the entry at 3.6.1; the ISO 18435-1:2009, 3.12 formulation is recorded as a regional variant with `status: similar` and the substantive variance (`systems or applications` → `entities`; `information` → `items` through an `interface`) named in the source block.

— **Field casing drift.** EN 18223:2026 specifies `lastUpdated`; IDTA 02099-1 implements `lastUpdate`; enumerated values drift in case between normative text (`model`, `batch`, `item`, `active`) and worked examples (`Model`, `Item`, `Active`). The drift is recorded here because it is reproduced in the terms of the corpus wherever example values are quoted.

— **Lifecycles.** Three incompatible lifecycle taxonomies coexist in the EN series: the identifier-granularity model of EN 18219:2026, the nine-phase model of EN 18239:2026 Annex A and the ISO 14040:2006 lineage used by other parts of the series. The affected entries retain their respective sources; no attempt is made in this document to impose a single lifecycle model.

— **Lookup systems.** EN 18219:2026 defines a resolver; EN 18222:2026 defines a registry API; the central EU registry is the object of a forthcoming implementing act; their relationships are unspecified in the corpus. The respective entries are kept distinct, and ISO/IEC 18975:2024 is recorded as the alignment target for the resolution model.

A.3 Relationship to further vocabularies

The entries of this document are aligned with the following further vocabularies:

— **ISO 5157:2023** (textiles — environmental aspects) is cited by the entries for textile value-chain concepts (textile collector, collection, textile product), recorded in the dataset as adopted sources;

— **ISO 59040** (circular economy — product circularity data sheet) is the alignment target for the material-flow entries (see 3.4 and 3.7) and for the transformation-algebra framework terms (3.11): the quantity carve-out and mass balance accounting entries are drafted to be compatible with its chain-of-custody concepts as standardized in ISO 22095:2020;

— **ISO 8601-1:2019** (date and time — representation for information interchange) and **ISO 80000-3** (quantities and units — Part 3: Space and time) are recorded as informative alignment targets for the time-stamping and units semantics of the framework.

— **UNTP** (UN Transparency Protocol) is the alignment target for the identity, carrier and resolution entries (3.2) and for the attestation triad recorded in the framework (3.11): the dataset's resolver and identifier concepts correspond to the UNTP product identifier and link-resolver model, and the verifiable-credential triad of UNTP corresponds to the framework's attestation concepts. These alignments are informative; they are recorded to prevent divergent redefinition as ISO/IEC JTC 5 work proceeds.

The machine-readable concept dataset from which the entries of 3.1 to 3.7 are converted is maintained by the UniDPP harmonization project and is published at https://glossarist.org/dpp/ (mirror https://unidpp.org/terminology/). The dataset's source-of-truth is 145 verified concept files; the audit findings of the companion audit report `/Users/mulgogi/src/isoiecjtc5/drafts/dataset-audit/AUDIT-REPORT.md` are addressed in this document by the merger, lineage citation and designation hygiene edits applied in 3.1 to 3.7.

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